

SLOWLY RESOLVING PNEUMONIA(FROM INFECTIOUS DISEASE)

Slowly Resolving Community-Acquired Pneumonia

- The natural history of adequately treated community-acquired pneumonia is usually that of prompt resolution.
- In healthy young military personnel recovering from pneumococcal pneumonia:-
 - ✓ Fever resolved after 2.5 days
 - ✓ Cough after 7.9 days, and
 - ✓ “Crackles” on physical examination of the chest by 8 days.
 - ✓ Normalization of the WBC count occurred by day 4.
- Similar results are found in more “typical” patients with community-acquired pneumonia, who were older with more comorbid illnesses.
- Vital sign stabilization usually occurred in 2 to 6 days, depending on the definition of normalization.
- Radiographic abnormalities may take 4 to 10 weeks to normalize, with younger (<50 years of age) patients resolving more quickly.
- Patients who fail to resolve their radiographic abnormalities by day 30, despite clearing of signs and symptoms, usually have significant underlying pulmonary disease such as **emphysema, chronic bronchitis, pneumoconiosis, or asthma.**
- **Slowly resolving pneumonia has been defined as less than 50% clearing of radiographic abnormalities at 2 weeks and less than complete clearing of abnormalities by 4 wks .**
- Host factors such as age greater than 50 years, a history of alcoholism, and the presence of significant underlying diseases (COPD, diabetes, congestive heart failure, asthma, malignancy) have been found to be associated with slow resolution of pneumonia.
- **Rate of resolution may also depend on the etiologic agent, with:-**
 - ✓ **S. pneumoniae usually requiring up to 6 weeks**
 - ✓ **Legionella usually requiring 2 to 6 months, and**
 - ✓ **Mycoplasma usually requiring several weeks.**
- When pneumonia appears to be slowly resolving, several explanations must be explored.
- The organism under treatment may have developed resistance, the wrong organism may have been treated, or pneumonia may be the incorrect diagnosis.
- A variety of clinical entities may mimic pneumonia, including bronchiolitis obliterans organizing pneumonia (BOOP), neoplasms, vasculitis, pulmonary hemorrhage, and eosinophilic pneumonia.

SLOWLY RESOLVING PNEUMONIA (FROM BABY NELSON)

Definition

- Persistent symptoms or radiographic abnormalities beyond the expected time course.

Expected course

* Clinical improvement:

- ✓ Uncomplicated bacterial pneumonia improves within 2- 4 days of antibiotics.

* Radiographic improvement

- ✓ Uncomplicated bacterial pneumonia improve within 4-6 wk
- ✓ Pneumococcal & chlamydial pneumonia require 1-3 mo
- ✓ Mycoplasma pneumoniae require 2 wk to 2 mo.
- ✓ Staphylococcal, legionella, and enteric gram-negative require 3-6 mo
- ✓ Viral pneumonia may require many months.

Causes

1. Inadequate therapy:- inappropriate antibiotic choice, dose, or poor compliance
2. Development of resistant organisms
3. Immunodeficiencies:- congenital or acquired
4. Obstructing endobronchial lesions e.g. foreign body, pulmonary sequestration
5. Decreased mucociliary clearance e.g. ciliary dyskinesia, cystic fibrosis
6. Nonbacterial causes: Viruses, fungi, parasites, and Mycobacteria
7. Noninfectious causes: hypersensitivity pneumonitis, sarcoidosis and Wegener Granulomatosis.

Workup

- Identify the offending organism: Blood, sputum, pleural, bronchoalveolar lavage, or lung biopsy can provide tissue for Gram stain and/or culture and thereby guide antibiotic choice.
- Chest CT scans (thin-cut and /or high-resolution)
- Anti-neutrophil cytoplasmic antibodies [ANCA] for Wegener granulomatosis.
- Flexible fiberoptic bronchoscopy
- Lung biopsy